

Phenology and canopy conductance limit the accuracy of 20 evapotranspiration models in predicting transpiration

Michael A. Forster^{a,b,c,*}, Tony D.H. Kim^c, Sylvie Kunz^c, Majed Abuseif^c, Vishal R Chulliparambil^c, Jannany Srichandra^c, Ruby N. Michael^c

^a Implexx Sense, PO BOX 285, Moorabbin, VIC 3189, Australia

^b Edaphic Scientific Pty Ltd, PO BOX 285, Moorabbin, VIC 3189, Australia

^c Cities Research Institute, Griffith University, 170 Kessels Road, Nathan, QLD 4111, Australia

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ABSTRACT

Transpiration is a fundamental biophysical process, directly measured in plants by dividing sap flow by total leaf area. Under non-limiting conditions, transpiration and reference evapotranspiration (ET_0) are hypothesized to be equal when ET_0 is normalized by the leaf area index of the reference crop, i.e., $LAI = 2.88$. Known as the $E_{2.88}$ model, it has only been tested with ET_0 derived from Penman-Monteith FAO56. Phenological influences on canopy conductance potentially decouple transpiration from atmospheric evaporative demand and lower the accuracy of $E_{2.88}$. This study tested the accuracy of 20 $E_{2.88}$ models in predicting apple (*Malus domestica* (Suckow) Borkh. var. Granny Smith) and pear (*Pyrus communis* L. var. Beurre Bosc Pear) transpiration over the 2020–2021 austral growing season. For apple, the Penman-Monteith ASCE-EWRI model had the highest predictive power with 7% error and $r^2 = 0.89$; whereas for pear the Valiantzas (2018, Eq. (7)) showed 2% error and $r^2 = 0.96$ evaluated via linear regression. Generally, models that included a humidity parameter had stronger predictive power than models excluding humidity. Yet, the predictive power of the $E_{2.88}$ models decreased considering the phenological phases for each crop. For apple, early and late season $E_{2.88}$ models underestimated transpiration by at least 27%. For pear, late season error increased to 7% as the $E_{2.88}$ models overestimated transpiration. Canopy conductance and the atmospheric decoupling factor were significantly greater in early and late season for apple and significantly lower in late season for pear. Therefore, phenology decreased the predictive power of the $E_{2.88}$ model in early and late season by decoupling physiological processes from atmospheric evaporative demand.

1. Introduction

Evapotranspiration is fundamental to the hydrologic cycle and has wide-ranging influences on climate, ecological interactions and the management of industrial and agricultural land (Jasechko et al., 2013; Landsberg et al., 2017). It is therefore essential that there is a continued improvement in the understanding of evapotranspiration dynamics via direct measurement and model comparisons.

Reference evapotranspiration (ET_0) is water vapor loss to the atmosphere from a 0.12 m tall uniform grass with extensive cover and unlimited by soil water (Doorenbos and Pruitt, 1977; Xiang et al., 2020). The most widely cited ET_0 model is the Penman-Monteith FAO56 (PM FAO) which combines atmospheric aerodynamics, solar radiation, temperature, and air moisture via a hypothetical reference crop (Allen et al., 1998). However, the PM FAO model is computationally complex

with a requirement for an array of sensors, yet it is the benchmark against which alternative, simplified ET_0 models are tested (e.g., Ahooghalandari et al., 2016; Djaman et al., 2015, 2019; Ndulue and Ranjan, 2021; Song et al., 2019; Valiantzas, 2006, 2013a, 2013b). Demonstrating the accuracy of simplified ET_0 models is important because of the opportunity for more widespread use of ET_0 in varied locations and applications such as irrigation, hydrology, water resource management, and urban infrastructure and building design (Abuseif et al., 2021; Ndulue and Ranjan, 2021; Shafieiyoun et al., 2020; Júnior et al., 2020).

Under strictly defined conditions, transpiration and ET_0 are equivalent (Pereira et al., 2006). The strict conditions include dense vegetation, unlimited soil water availability or nutrition, and limited external stress such as parasitism or predation (Pereira et al., 2006). The transpiration and ET_0 equivalency model is called the $E_{2.88}$ model because it

* Corresponding author.

E-mail address: michael@edaphic.com.au (M.A. Forster).

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is the PM FAO model divided by the leaf area index of the hypothetical reference 0.12 m height grass crop which equals 2.88 (Allen et al., 1989; Pereira et al., 2006). That is, the $E_{2.88}$ model is ET_o normalized by leaf area of the reference crop. In a recent meta-analysis, the accuracy of the $E_{2.88}$ model was demonstrated across 13 woody tree species (Forster, 2021). Currently, the $E_{2.88}$ model has only been derived from the PM FAO model. Due to the lack of studies testing the $E_{2.88}$ model, it is not known whether alternative ET_o models display similar predictive power. A confirmation of the accuracy of alternative ET_o models may provide simpler and lower cost options to predict and model transpiration.

Transpiration (E) is directly measured by dividing sap flow (Q) by total tree leaf area (A_L). Although Q , by definition, occurs within plants, many studies have demonstrated a strong positive linear correlation between Q and ET_o (e.g., Deng et al., 2017; Forster and Englefield, 2021; Rana et al., 2005; Wang et al., 2019; Zhao and Zhao, 2015). As the definition of E is Q normalized by A_L (i.e., $E = Q/A_L$), and the $E_{2.88}$ model is ET_o normalized by A_L of the reference crop, it is hypothesized that E and $E_{2.88}$ are equivalent (i.e., $Q/A_L = ET_o/2.88$). Therefore, the relationship between E and $E_{2.88}$ can be directly tested through the measurement of Q , A_L and ET_o .

An assumption of the $E_{2.88}$ model is that Q scales with changes in A_L . That is, if there is an increase or decrease in A_L , such as early season leaf expansion or late season defoliation, there will be a proportional change in Q such that Q/A_L always equals $ET_o/2.88$. A scaling relationship between Q and A_L has long been established (Čermák et al., 2004). However, phenology may lead to variability in plant water relations due to changes in xylem hydraulic conductivity, stomatal control, and other processes related to leaf development and senescence (Forster, 2014; Tu et al., 2019; Yu et al., 2018; Zhao et al., 2019). Consequently, phenology may decouple the relationship between Q , A_L and ET_o with the effect of lowering the predictive power of the $E_{2.88}$ model. Therefore, a further exploration of the potential for phenological processes to lower the predictive power of the $E_{2.88}$ model is required.

The decoupling of transpiration from atmospheric evaporative demand can be quantified via canopy stomatal conductance measurements (Jarvis and McNaughton, 1986; Lu et al., 2003). Canopy stomatal conductance is the ease to which water vapor can transfer from vegetation to the atmosphere and is estimated from an inverse PM FAO model (Monteith and Unsworth, 1990). A decoupling factor can also be estimated from measurements of E and the PM FAO model (Jarvis and McNaughton, 1986). An increase in the decoupling factor correlated with phenology suggests that transpiration is weakly linked with atmospheric evaporative demand. Therefore, it is hypothesised that $E_{2.88}$ has a lower predictive power for E when canopy conductance is increasingly decoupled from atmospheric biophysical controls.

This study measured Q and A_L in two deciduous tree species in an intensively managed agricultural system under non-limiting conditions to test the following hypotheses: Q and ET_o estimated from 20 models are equivalent when normalized via leaf area (i.e. $E = E_{2.88}$); phenology influences the predictive power of $E_{2.88}$ models; and the decrease in $E_{2.88}$ predictive power is caused by an increase in the decoupling factor.

2. Methods

2.1. Site and species description

The study was conducted over the austral growing season from October 2020 to June 2021. Apple, *Malus domestica* (Suckow) Borkh. (var. Granny Smith); and pear, *Pyrus communis* L. (var. Beurre Bosc Pear) trees were monitored in an established orchard in Yarra Valley, Victoria, Australia (37.73° S, 145.45° E). The site elevation was ~90 m a.s.l., consisting of low rises and swales, with long-term mean maximum / minimum annual temperature of 20.5 / 7.4°C and mean annual rainfall of 732 mm. The soils were comprised of a silty clay or clay loam to a depth of approximately 300 to 500 mm below which there was a gradation to a semi-impermeable sub-soil layer.

Table 1

A list of parameters, definitions and units of measurement used for the calculation of sap flow, transpiration and evapotranspiration.

Parameter	Definition	Units
A_L	Total canopy leaf area.	m ²
A_S	Sapwood area.	m ²
c_d	Dry wood specific heat capacity.	J kg ⁻¹ K ⁻¹
c_w	Sap solution specific heat capacity.	J kg ⁻¹ K ⁻¹
C_d	Numerator constant for the PM ASCE-EWRI model. For a short reference crop, day is 0.24 and night is 0.96.	Dimensionless
C_n	Denominator constant for the PM ASCE-EWRI model. C_n equals 37 for a short reference crop.	Dimensionless
C_p	Dry air specific heat capacity at constant pressure.	J kg ⁻¹ K ⁻¹
D	Saturated vapor pressure deficit.	kPa
D_{avg}	Daily average saturated vapor pressure deficit.	kPa
E	Transpiration.	mm hr ⁻¹ or mm d ⁻¹
ET_o	Reference evapotranspiration.	mm hr ⁻¹ or mm d ⁻¹
G_a	Aerodynamic conductance.	m s ⁻¹
G_c	Canopy conductance.	m s ⁻¹
G_t	Canopy conductance of the average stomata.	m s ⁻¹
J	Sap flux density.	m ³ m ⁻² s ⁻¹
LAI	Leaf area index.	Dimensionless
m_c	Sapwood gravimetric water content.	m ³ m ⁻³
P	Precipitation.	mm
Q	Total tree volumetric sap flow.	L hr ⁻¹ or L d ⁻¹
RH_{avg}	Daily average of relative humidity.	Dimensionless
R_n	Net solar radiation.	MJ m ⁻² d ⁻¹
R_s	Incoming solar radiation.	MJ m ⁻² d ⁻¹
T	Air temperature.	°C
T_{avg}	Daily average air temperature.	°C
$T_{avg,K}$	Daily average air temperature in Kelvin.	K
T_{max}	Maximum daily air temperature.	°C
T_{min}	Minimum daily air temperature.	°C
u_2	Wind speed at 2 m height.	m s ⁻¹
$u_{2,avg}$	Daily average wind speed at 2 m height.	m s ⁻¹
V_c	Corrected heat velocity.	m s ⁻¹
W_{aero}	0.78 ($RH_{avg} > 0.65$); 1.067 ($RH_{avg} \leq 0.65$).	Dimensionless
Z	Elevation above sea level of weather station.	m
α	Albedo factor (0.23 or 0.25).	Dimensionless
γ	Psychrometric constant.	kPa °C ⁻¹
Δ	Slope of the saturation vapour pressure curve at ambient air temperature.	kPa °C ⁻¹
λ	Latent heat of vaporization of water.	MJ kg ⁻¹
ρ	Density of dry air.	kg m ⁻³
ρ_d	Density of dry sapwood.	kg m ⁻³
ρ_w	Density of water.	kg m ⁻³
ϕ	Latitude of weather station.	Radians
Ψ	Soil water potential.	kPa
Ω	Canopy conductance decoupling factor.	Dimensionless

Apple trees were approximately 15 years of age, 4.5 to 5.5 m in height, 0.2 m in basal diameter with a sapwood specific density of 557 kg m⁻³ and sapwood depth of 0.05 m. Within row and interrow canopy radius was approximately 1.7 m and 2.0 m, respectively, with canopies overlapping within rows. Tree spacing was 2.5 m and interrow spacing was 5.0 m. Five individual apple trees were measured across two blocks approximately 500 m distance. Within the first block, a row was randomly selected and then three trees within that row were measured. Another row was randomly selected in the second block and two trees within that row were measured.

Pear trees were approximately 15 years of age, 5 m in height, 0.25 m in basal diameter with a sapwood specific density of 566 kg m⁻³ and sapwood depth of 0.065 m. Within row and interrow canopy radius was approximately 1.7 m and 2.2 m, respectively, with canopies overlapping within rows. Tree spacing was 3.0 m and interrow spacing was 5.0 m. Three individual pear species were measured for sap flow within a single block where a random row was selected within the block for measurements.

Soil water potential was maintained at a high level (>-50 kPa) with frequent irrigation and was monitored with TEROS 21 sensors (METER

Table 2

A list of the 20-reference evapotranspiration (ET_o) models used in this study. Parameters and their definition and units of measurement are provided in Table 1.

ET_o Model Name	Abbrev.	ET_o Equation	Reference
Ahooghalandari et al Eq. (11)	Ahoogh_1	$ET_o = 0.139R_n + 0.182T_{avg} (1 - RH_{avg}) - 1.55$	Ahooghalandari et al (2016)
Ahooghalandari et al Eq. (12)	Ahoogh_2	$ET_o = 0.151R_n + 0.139T_{max} (1 - RH_{avg}) - 1.95$	Ahooghalandari et al (2016)
Allen	Allen	$ET_o = 0.001183R_s (T_{avg} + 20) (T_{max} - T_{min})^{0.4}$	Allen (1993)
Capiro	Capiro	$ET_o = R_s (0.01092708T + 0.0060706)$	Capiro (1974)
Hargreaves	Harg	$ET_o = 0.0055R_s (T_{avg} + 17.8)$	Hargreaves (1975)
Hargreaves-Samani	HS	$ET_o = 0.000938R_n (T_{avg} + 17.8) (T_{max} - T_{min})^{0.5}$	Hargreaves and Samani (1985)
Penman-Monteith ASCE-EWRI	PM ASCE	$ET_o = \frac{0.408\Delta R_n + C_n \gamma Du_2 / (T + 273)}{\Delta + \gamma(1 + C_d u_2)}$	Allen et al (2005)
Penman-Monteith FAO56	PM FAO	$ET_o = \frac{0.408\Delta R_n + 900Du_2 / (T + 273)}{\Delta + \gamma(1 + 0.34u_2)}$	Allen et al (1998)
Priestley-Taylor	PT	$ET_o = 1.26(\Delta / [\Delta + \gamma]) (R_n / \lambda)$	Priestley and Taylor (1972)
Ritchie	Ritchie	$ET_o = (\Delta / [\Delta + \gamma]) (0.408R_n)$	Ritchie (1972)
Romanenko	Roman	$ET_o = 4.5(1 + [T_{avg} / 25])^2 (1 - RH_{avg})$	Romanenko (1961)
Turc	Turc	$ET_o = 0.013 (23.88R_s + 50) (T_{avg} / [T_{avg} + 15])$	Turc (1961)
Valiantzas (2006) Eq. (38)	Val_1	$ET_o \approx 0.051 (1 - \alpha) R_s (T_{avg} + 9.5)^{0.5} - 2.4 (R_s / R_a)^2 + 0.048 (T_{avg} + 20) (1 - RH_{avg}) (0.5 + 0.536u_{2,avg}) + 0.00012Z$	Valiantzas (2006)
Valiantzas (2006) Eq. (39)	Val_2	$ET_o \approx 0.038R_s (T_{avg} + 9.5)^{0.5} - 2.4 (R_s / R_a)^2 + 0.075 (T_{avg} + 20) (1 - RH_{avg})$	Valiantzas (2006)
Valiantzas (2012)	Val_3	$ET_o \approx 0.0393R_s (T_{avg} + 9.5)^{0.5} - 2.4 (R_s / R_a) - 0.024 (T_{avg} + 20) (1 - RH_{avg}) + 0.066W_{aero} (T_{avg} + 20) (1 - RH_{avg}) u_{2,avg}^{0.6}$	Valiantzas (2012)
Valiantzas (2013a)	Val_4	$ET_o \approx 0.0393R_s (T_{avg} + 9.5)^{0.5} - 0.19R_s^{0.6} \phi^{0.15} + 0.0061 (T_{avg} + 20) (1.12T_{avg} - T_{min} - 2)^{0.7}$	Valiantzas (2013a)
Valiantzas (2013b)	Val_5	$ET_o \approx 0.0393 (T_{avg} + 9.5)^{0.5} (0.408R_s - 0.07752R_s^{0.6}) \phi^{0.15} + 0.0061 (T_{avg} + 20) (1.12T_{avg} - T_{min} - 2)^{0.7}$	Valiantzas (2013b)
Valiantzas (2013b) Eq. (34)	Val_6	$ET_o \approx 0.00668R_a (T_{avg} + 9.5)^{0.5} (T_{max} - T_{min}) - 0.0696 (T_{max} - T_{min}) - 0.024 (T_{avg} + 20) (1 - RH_{avg}) - 0.0045R_a (T_{max} - T_{min})^{0.5} + 0.0984 (T_{avg} + 17) (1.03 + 0.00055 (T_{max} - T_{min})^2 - RH_{avg})$	Valiantzas (2013b)
Valiantzas (2018) Eq. (7)	Val_7	$ET_o \approx 0.0118 (1 - RH_{avg})^{0.2} (T_{max} - T_{min})^{0.3} (R_a [T_{avg} + 10]^{0.5} - 40) + 0.1(T_{avg} + 20) (1 - RH_{avg})$	Valiantzas (2018)
Valiantzas (2018) Eq. (8)	Val_8	$ET_o \approx 0.0118 (1 - RH_{avg})^{0.2} (T_{max} - T_{min})^{0.3} (R_a [T_{avg} + 10]^{0.5} - 40) + 0.1(T_{avg} + 20) (1 - RH_{avg}) (u_{2,avg} / 2)^{0.6}$	Valiantzas (2018)

Group, Pullman, USA), installed at 0.2 m and 0.4 m depth. Across the growing season and measurement campaign, irrigation occurred every two to three days via a drip irrigation system, dripper spacing of 2.5 m, delivery rate of 8 L h^{-1} over a 6-hour period. Apple and pear trees were grown under a permanently installed white canopy (bird and hail protection) net with 0.002 m diameter holes. The canopy net was constructed at a height of 6 m and held aloft by posts installed approximately 10 to 15 m apart.

2.2. ET_o Models

ET_o models were selected from the review papers by Djaman et al. (2019), Liu et al. (2017), Song et al. (2019) and Xiang et al. (2020). ET_o models that were valid on a daily time scale or consisted of hourly time scales that could be summed into 24 periods, were used in this study. ET_o models that were based on 5-day, monthly or longer time steps were excluded. ET_o models that were based on site specific, localized or empirical calibrations (e.g., Dorji et al., 2016; Droogers and Allen, 2002) were excluded. Table 1 is a list of parameters, and their definition and units of measurement, used in the ET_o models. Table 2 is a list of the 20 ET_o models selected for this study.

The Penman-Monteith FAO56 and ASCE-EWRI models included a soil heat flux parameter for the calculation of ET_o however instrumentation for its measurement was unavailable in this study. Allen et al. (1998) and Wallace (pers. comm.) noted that soil heat flux could be disregarded because it has a minimal influence on ET_o calculations. Additionally, when the focus is on daily ET_o calculations, it is recommended to estimate G from net radiation for hourly values.

2.3. Meteorological and weather measurements

An ATMOS 41 (METER Group, Pullman, USA) weather station was installed centrally within the orchard with a minimal clearance of 10 m radius beneath the same white canopy net as the apple and pear trees. In

this position, the weather station sensors, solar radiation ($\pm 5 \%$), air temperature ($\pm 0.6^\circ\text{C}$), relative humidity ($\pm 2 \%$), and wind speed ($\pm 0.3 \text{ m s}^{-1}$ or $\pm 3\%$ of measurement), were located 2 m above a uniform grass layer maintained at approximately 0.12 m height.

2.4. Sap flow measurements

Sap flow was measured using an Implexx Sap Flow Sensor (Implexx Sense, Melbourne, Australia) consisting of three stainless steel needles, 0.0013 diameter and spaced at an equidistance of 0.006 m, installed on the north-facing stem, at approximately 0.3 m height from the base of the trunk, with the outer downstream and upstream thermistor positioned at exactly 0.005 m sapwood depth. The downstream and upstream needles contained two negative temperature coefficient (NTC) thermistors located at 0.01 and 0.02 m radial depths. Thermistor measurement accuracy was $\pm 0.001^\circ\text{C}$ with a time resolution of 200 milliseconds. The central needle contained a nichrome, of 39 ohms resistance, heater element. A 3-second heat pulse delivered an approximate pulse energy of 400 J m^{-1} from the heater element per measurement cycle. The sensors were insulated with aluminium foil.

Sap flow was estimated via the Dual Method Approach (DMA). Extensive details on the DMA, including theory and equations, were outlined in Forster (2019, 2020). Briefly, the DMA is a heat pulse method derived from the theory of conduction and convection of heat in porous materials, that combines the slow rates of flow from the heat ratio method and fast rates of flow from the Tmax method.

Using DMA, the switch between slow and fast methods was determined via the Peclet number which is the ratio of conduction to convection: when the Peclet number was ≤ 1 , the slow flow method was chosen; otherwise, the fast flow method was chosen. The data output from the DMA method was heat velocity (m s^{-1}) converted to corrected heat velocity (m s^{-1}), to account for the effects of xylem wounding, following the methods of Forster (2020).

Corrected heat velocity (V_c) was subsequently converted to sap flux

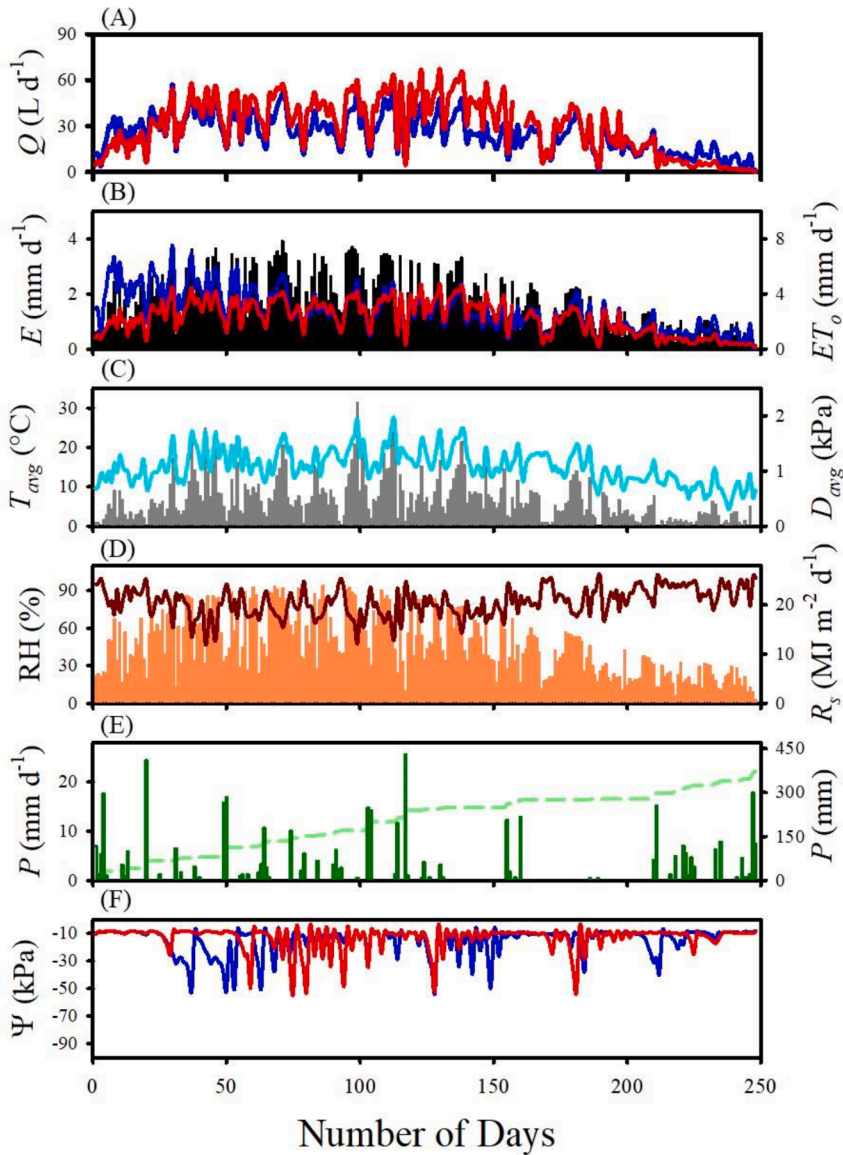


Fig. 1. A summary of weather parameters at the study site from early October 2020 to mid-June 2021. Detailed parameter definitions are listed in Table 1. (A) Daily sap flow (Q) for apple (blue) and pear (red) trees. (B) Daily transpiration (E) for apple (blue) and pear (red) trees and daily evapotranspiration (black, ET_o). (C) Daily average air temperature (T_{avg} ; teal line) and vapor pressure deficit (D ; grey bars). (D) Daily minimum relative humidity (RH ; brown line) and solar radiation (R_s ; orange bars). (E) Daily precipitation (P ; green bars) and cumulative precipitation (P ; dashed green line). (F) Soil water potential (Ψ) for apple (blue) and pear (red) trees. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

density (J , $\text{m}^3 \text{m}^{-2} \text{s}^{-1}$, Barrett et al., 1995):

$$J = V_c \rho_d (c_d + m_c c_w) / \rho_w c_w \quad (1)$$

where ρ_d is sapwood density (kg m^{-3}), c_d is dry wood specific heat capacity ($1200 \text{ J kg}^{-1} \text{K}^{-1}$; Becker and Edwards, 1999), m_c is sapwood gravimetric moisture content (kg kg^{-1}), c_w is sap solution specific heat capacity ($4182 \text{ J kg}^{-1} \text{K}^{-1}$; Becker and Edwards, 1999), and ρ_w is the density of sap (1000 kg m^{-3} ; Becker and Edwards, 1999). Sapwood cores were used to measure ρ_d and m_c , thermal diffusivity following methods outlined by Forster (2019). A stem borer (Haglöf, Sweden) was used to destructively sample 0.005 m diameter cores from apple and pear trees proximate to the instrumented trees. Cores were sampled following the installation of sap flow sensors in late September, 2021. Only the outer 0.01 m of the sapwood core was retained for ρ_d and m_c measurements.

Total tree sap flow (Q , kg or L) was calculated as J multiplied by sapwood area (A_s , m^2). A_s was estimated from the sapwood cores sampled from the trees proximate to the instrumented trees. An allometric relationship between sapwood depth, sapwood area and trunk diameter were determined via the sapwood cores. This relationship was then used to estimate A_s of the instrumented trees. Sapwood depth (radius) was easily measured on the sample cores because there was a

clear visual demarcation between sapwood and heartwood. As noted above, sapwood depth for apple and pear trees average 0.05 and 0.065 m, respectively. Consequently, A_s was divided into three annuli, outer, inner and non-measured, to correspond with the measurements from the sap flow sensor. The outer annulus was the depth between bark and 0.01 m sapwood radius and the inner annulus was between 0.01 and 0.02 m sapwood radius. The non-measured annulus was between 0.02 and 0.05 m radius for apple and 0.02 and 0.065 m radius for pear. Therefore, Q was calculated as:

$$Q = (J_{outer} A_{s_outer}) + (J_{inner} A_{s_inner}) + (J_{non-measured} A_{s_non-measured}), \quad (2)$$

where $J_{non-measured}$ was estimated as $0.5 * J_{inner}$ following Pausch et al. (2000).

Lastly, sap flow data were corrected for probe misalignment. Sap flow measurements via the DMA assume the distance between the heater source and temperature sensor is exactly 0.006 m. In practice, the distance varies from 0.006 m because of variations in installation drilling techniques as well as irregular xylem pathways within xylem. Probe misalignment can be determined by examining data during periods of expected zero sap flow. This includes nocturnal periods, where the soil moisture is at or near saturation, and atmospheric conditions are

saturated (i.e. relative humidity is $\sim 100\%$ or vapour pressure deficit is ~ 0 kPa). Where required, sap flow data were adjusted to zero based on collected data.

2.5. Leaf area measurements and phenological periods

Total leaf area (A_L , m^2) for instrumented apple and pear trees was estimated by multiplying leaf area index (LAI) by canopy projection area. LAI was estimated via averaging measurements from two, independent light interception methods. A ceptometer (AccuPAR, model LP-80, METER Group, Pullman, USA) was used to average 20 below canopy measurements that were taken at random positions beneath each tree approximately 0.5 m distance from the trunk. The second LAI method was digital imagery where images at four locations (north, south, east, west) beneath the canopy approximately 0.5 m distance from the trunk were converted to LAI via a MATLAB based digital photography algorithm developed by Fuentes et al. (2008). Results from the ceptometer and digital photography algorithm were similar therefore the average of the two methods was used. Canopy projection area was assumed to be circular after averaging eight radii measurements, between the trunk and outer canopy, around the circumference of a typical tree. LAI and canopy projection area were measured at least every 2 weeks during the season and then more frequently at weekly intervals during the late season.

Apple and pear phenology was divided into three periods: early, mid and late season. The three periods were quantified by the relative change in A_L . The early, mid and late season was demarcated by an increase in A_L , a stable A_L , and a decreasing A_L , respectively. These periods denote when the canopy and leaves were developing, mature and undergoing senescence.

2.6. Canopy conductance and decoupling factor

Two methods to estimate canopy conductance were used in this study. The first method is an inverse of the Penman-Monteith FAO56 evapotranspiration equation (see Table 2; Monteith and Unsworth, 1990):

$$G_c = E\lambda G_a / \Delta R_n + \rho C_p D_{avg} G_a - E\lambda(\Delta + \gamma), \quad (3)$$

where G_c , E , λ , G_a , Δ , ρ , C_p , D_{avg} , and γ are defined in Table 1.

The second method is an estimate of the canopy conductance and is the “average” stomata in the canopy (Köstner et al., 1992):

$$G_t = (G_s T_{avg,K}) (E / D_{avg}), \quad (4)$$

where G_b , G_s , $T_{avg,K}$, E and D_{avg} are defined in Table 1.

The decoupling factor, Ω , was defined by Jarvis and McNaughton (1986) as:

$$\Omega = (\Delta / \gamma + 1) / (\Delta / \gamma + 1 + G_a / G_c), \quad (5)$$

where Δ , γ , G_a and G_c are defined in Table 1.

2.7. Data analysis and time steps

All data analyses were performed over 24-hour daily time steps. Data were collected at 15 minute or hourly time intervals and summed to daily totals. A 24-hour period was considered as 5AM to 4AM, rather than midnight to midnight, because daily tree physiological processes occur between sunrise to sunrise.

Linear regression analysis was used to test the hypothesis that E is predicted from $E_{2.88}$ (i.e., $ET_o / 2.88$). All data were untransformed because the analysis tested whether E was linearly related to $E_{2.88}$ and a log or other transformation would bias this assumption. Slopes of the linear regression curve were fitted through the origin assuming there was a 1:1 relationship between E and $E_{2.88}$. A slope and r^2 of 1 indicates a

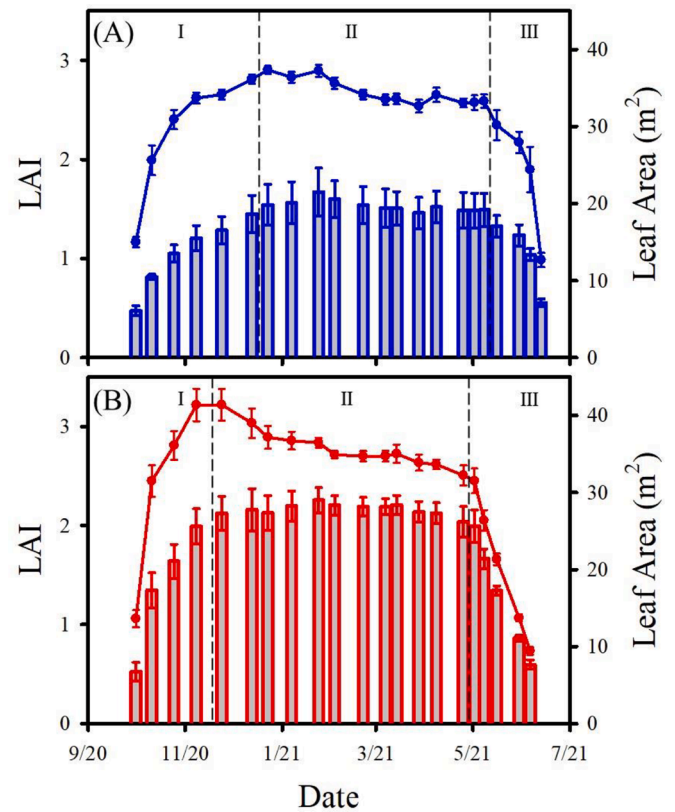


Fig. 2. Leaf area index (LAI) and total tree leaf area for (A) apple and (B) pear trees over the entire measurement period. Data are mean $\pm$ standard error. The line and circles are LAI and the bar charts are leaf area. The phenological periods are indicated by the dashed lines where I is early, II is mid-, and III is late season.

perfect relationship, whereas a slope less than 1 indicates $E_{2.88}$ overestimates E and a slope greater than 1 indicates that E is greater than predicted $E_{2.88}$. The accuracy of each ET_o model were further assessed via a root mean square error (RMSE) test where smaller values indicate a model has greater predictive power.

One-way ANOVA was used to test the hypothesis that G_c , G_t or Ω varied with phenology. Data were checked for normality and transformed where required. Following a significant difference in the ANOVA F -test, a post hoc analysis using Tukey's HSD tested the differences between early, mid and late season.

3. Results

Apple and pear season was between October 2020 and June 2021. Sap flow (Q) reached a maximum average of $54.7 (\pm 9.6 \text{ S.E.}) \text{ L d}^{-1}$ per apple tree and $66.2 (\pm 8.5 \text{ S.E.}) \text{ L d}^{-1}$ per pear tree (Fig. 1). Maximum seasonal Q occurred in mid to late November 2020 (late spring) in apples and early February 2021 (mid-summer) for pears. Transpiration (E) reached a maximum average of $3.6 (\pm 0.3 \text{ S.E.}) \text{ mm d}^{-1}$ per apple tree and $2.3 (\pm 0.2 \text{ S.E.}) \text{ mm d}^{-1}$ per pear tree corresponding to late spring to mid-summer for apple and pears, respectively (Fig. 1). Daily ET_o , estimated via the Penman-Monteith FAO56 model, reached a maximum of 7.8 mm d^{-1} in mid-December 2021 (early summer; Fig. 1). A summary of other weather variables over the measurement period is displayed in Fig. 1.

Fig. 2 shows the results for LAI and total canopy leaf area for apple and pear trees. For apple trees, LAI increased between early October 2020 and reached a maximum in late December 2020. Thereafter, LAI exhibited a steady decline until mid-May 2021 when LAI decreased during canopy senescence. Leaf area was more stable between late-

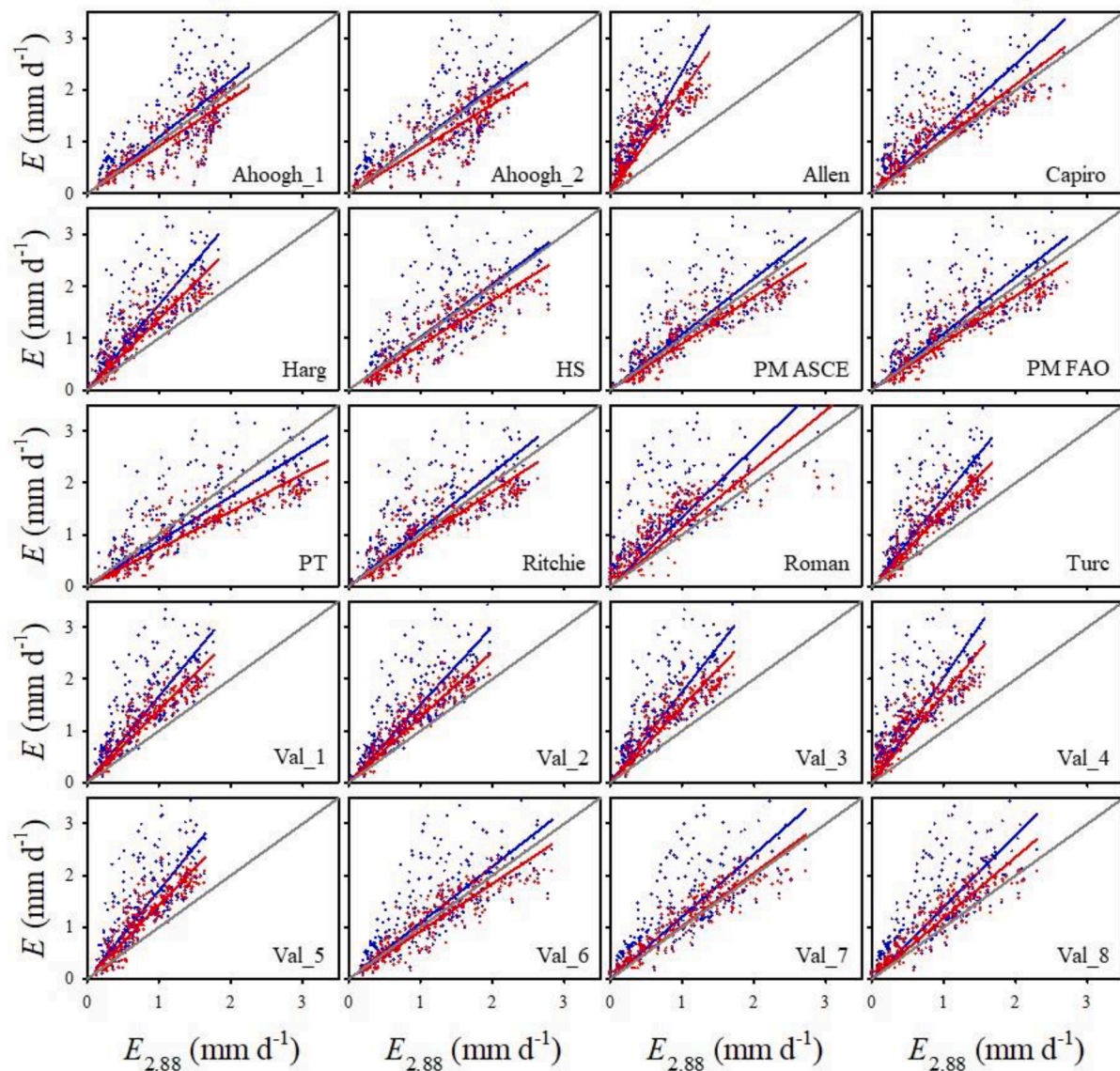


Fig. 3. The predictive power of 20 $E_{2.88}$ models for E in apple trees (blue) and pear trees (red). The $E_{2.88}$ model abbreviation is labelled on each graph and defined in Table 2. The grey line is the 1:1 line of unit. Statistical results are presented in Tables 3 and 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

December and mid-May corresponding to an increase in canopy projection area. The pear trees showed a similar pattern in LAI to the apple trees except maximum LAI was reached earlier in approximately mid-November 2021. Leaf area was also stable during the mid-season due to an increase in canopy projection area.

The phenological periods are indicated for apple and pear trees in Fig. 2. For apple trees, early, mid and late season phenological periods corresponded to 5th October 2020 to 22nd December 2020, 23rd December 2020 to 10th May 2021, and 11th May 2021 to 9th June 2021, respectively. For pear trees, early, mid and late season phenological periods corresponded to 5th October 2020 to 26th November 2020, 27th November 2020 to 6th May 2021, and 7th May 2021 to 9th June 2021, respectively.

The results of the linear regression for apple and pear E for all 20 $E_{2.88}$ models over the entire measurement period are displayed in Fig. 3. A summary of the linear regression analysis for all models is tabulated in Table 3 for apple and Table 4 for pear. For apple, all $E_{2.88}$ models underestimated E except for the Priestley-Taylor model. The model with the best predictive power (i.e. lowest RMSE value) for apple was the

Penman-Monteith ASCE-EWRI model. For pear, the results were mixed with approximately half of the $E_{2.88}$ models over or underestimating E . The model with the best predictive power for pear was the Valiantzas (2018) Eq. (7) model.

The influence of phenology on the predictive power of the $E_{2.88}$ models are shown in Fig. 4 and Table 3 for apple, and Fig. 5 and Table 4 for pear. For apple, all $E_{2.88}$ models, except for the Priestley-Taylor model, underestimated E during the early and late seasons. During the mid-season, the $E_{2.88}$ models displayed stronger predictive power with the Valiantzas (2018) Eq. (7) model having the lowest RMSE. In contrast, many of the $E_{2.88}$ models showed strong predictive power during the early and mid-season with the Valiantzas (2018) Eq. (7) model also having the lowest RMSE. For pear trees during the late season, generally there was a decrease in predictive power where several $E_{2.88}$ models overestimated pear E .

Canopy conductance and decoupling factor results are displayed in Fig. 6. Note that Fig. 6 displays averages and standard errors yet statistical analysis was performed on transformed data to meet the assumptions of normality. For apple, there was a significant difference in

Table 3

The predictive power of the $E_{2.88}$ model, derived from 20 different ET_o models, for estimating E in apple trees. RMSE is the root mean square error where smaller values equal a model with greater predictive power. Bold values indicate the model with the best predictive power during that phenological period. All linear regressions were significant at $p < 0.05$. The definition and equation for each model is presented in Table 2.

Model	All Season			Early Season			Mid-Season			Late Season		
	Slope	r^2	RMSE	Slope	r^2	RMSE	Slope	r^2	RMSE	Slope	r^2	RMSE
Ahoogh_1	1.09	0.87	0.59	1.28	0.89	0.85	0.93	0.90	0.43	2.21	0.93	0.37
Ahoogh_2	1.02	0.87	0.58	1.20	0.89	0.80	0.87	0.90	0.45	2.67	0.95	0.41
Allen	2.35	0.85	1.04	2.77	0.85	1.53	2.02	0.91	0.74	3.48	0.81	0.49
Capiro	1.25	0.85	0.67	1.51	0.85	1.07	1.06	0.93	0.36	2.09	0.86	0.38
Harg	1.64	0.88	0.79	1.94	0.89	1.22	1.40	0.94	0.49	2.21	0.88	0.39
HS	1.02	0.87	0.57	1.24	0.88	0.85	0.86	0.94	0.39	1.36	0.87	0.28
PM ASCE	1.07	0.89	0.53	1.27	0.90	0.82	0.92	0.94	0.33	1.26	0.88	0.25
PM FAO	1.09	0.89	0.54	1.29	0.90	0.84	0.93	0.94	0.33	1.29	0.88	0.26
PT	0.87	0.88	0.59	1.03	0.89	0.72	0.74	0.94	0.56	1.01	0.85	0.25
Ritchie	1.09	0.88	0.55	1.29	0.89	0.85	0.94	0.94	0.35	1.27	0.85	0.27
Roman	1.34	0.81	0.78	1.67	0.79	1.26	1.13	0.91	0.41	1.56	0.86	0.31
Turc	1.72	0.90	0.81	2.05	0.90	1.25	1.46	0.95	0.50	2.07	0.92	0.36
Val_1	1.67	0.90	0.78	1.95	0.91	1.20	1.43	0.95	0.49	2.55	0.96	0.40
Val_2	1.51	0.90	0.71	1.79	0.91	1.12	1.28	0.96	0.39	2.68	0.97	0.36
Val_3	1.75	0.89	0.82	2.03	0.90	1.24	1.51	0.94	0.54	2.84	0.93	0.42
Val_4	2.02	0.86	0.95	2.38	0.86	1.42	1.74	0.93	0.64	3.40	0.84	0.48
Val_5	1.70	0.89	0.81	2.09	0.90	1.26	1.44	0.96	0.48	1.66	0.85	0.33
Val_6	1.09	0.88	0.57	1.31	0.88	0.88	0.92	0.94	0.33	1.90	0.96	0.32
Val_7	1.20	0.88	0.61	1.46	0.88	0.97	1.02	0.94	0.31	2.37	0.97	0.37
Val_8	1.39	0.87	0.70	1.65	0.88	1.09	1.18	0.94	0.39	3.12	0.98	0.43

Table 4

The predictive power of the $E_{2.88}$ model, derived from 20 different ET_o models, for estimating E in pear trees. RMSE is the root mean square error where smaller values equal a model with greater predictive power. Bold values indicate the model with the best predictive power during that phenological period. All linear regressions were significant at $p < 0.05$. The definition and equation for each model is presented in Table 2.

Model	All Season			Early Season			Mid-Season			Late Season		
	Slope	r^2	RMSE	Slope	r^2	RMSE	Slope	r^2	RMSE	Slope	r^2	RMSE
Ahoogh_1	0.91	0.93	0.36	0.88	0.94	0.38	0.93	0.93	0.39	1.00	0.96	0.06
Ahoogh_2	0.86	0.94	0.38	0.83	0.95	0.41	0.87	0.93	0.41	1.19	0.97	0.07
Allen	1.98	0.91	0.71	2.03	0.93	0.75	1.96	0.91	0.77	1.65	0.83	0.16
Capiro	1.05	0.93	0.36	1.10	0.94	0.35	1.04	0.92	0.39	0.97	0.87	0.11
Harg	1.37	0.94	0.45	1.37	0.96	0.44	1.38	0.94	0.50	1.03	0.88	0.10
HS	0.87	0.95	0.35	0.89	0.96	0.33	0.86	0.95	0.38	0.64	0.93	0.18
PM ASCE	0.90	0.95	0.32	0.90	0.97	0.28	0.90	0.95	0.35	0.59	0.88	0.22
PM FAO	0.91	0.95	0.32	0.91	0.97	0.27	0.91	0.95	0.35	0.61	0.88	0.21
PT	0.72	0.94	0.57	0.72	0.96	0.57	0.73	0.94	0.61	0.48	0.87	0.32
Ritchie	0.91	0.94	0.33	0.91	0.96	0.30	0.92	0.94	0.36	0.60	0.87	0.21
Roman	1.13	0.88	0.46	1.09	0.84	0.55	1.16	0.90	0.47	0.67	0.68	0.21
Turc	1.43	0.95	0.47	1.43	0.97	0.46	1.44	0.95	0.52	0.96	0.93	0.08
Val_1	1.39	0.96	0.44	1.37	0.98	0.41	1.40	0.96	0.49	1.17	0.93	0.09
Val_2	1.26	0.97	0.35	1.25	0.98	0.33	1.27	0.96	0.39	1.04	0.93	0.08
Val_3	1.46	0.95	0.48	1.42	0.98	0.44	1.48	0.95	0.54	1.31	0.93	0.10
Val_4	1.70	0.93	0.61	1.71	0.94	0.63	1.70	0.93	0.67	1.55	0.83	0.16
Val_5	1.43	0.95	0.47	1.47	0.97	0.48	1.43	0.95	0.51	0.79	0.89	0.13
Val_6	0.92	0.96	0.29	0.93	0.96	0.28	0.92	0.95	0.32	0.86	0.95	0.08
Val_7	1.02	0.96	0.25	1.03	0.97	0.23	1.02	0.96	0.28	1.07	0.91	0.09
Val_8	1.18	0.96	0.32	1.18	0.97	0.30	1.17	0.95	0.36	1.40	0.93	0.11

G_c ($F = 120.153$, $p < 0.001$, log-transformed data), G_t ($F = 56.876$, $p < 0.001$, log-transformed data), and Ω ($F = 77.681$, $p < 0.001$, untransformed data). The Tukey HSD post hoc analysis showed that early and late season G_c , G_t and Ω was significantly greater than mid-season. For pear, there was a significant difference in G_c ($F = 12.005$, $p < 0.001$, log-transformed data) and Ω ($F = 6.602$, $p = 0.001$, untransformed data), but no significant difference in G_t ($F = 2.720$, $p = 0.066$, log-transformed data). The Tukey HSD post hoc analysis showed that late season G_c and Ω were significantly less than early and mid-season values for pear.

4. Discussion

Phenology and canopy conductance decreased the predictive power of 20 $E_{2.88}$ models in estimating apple and pear E . Generally, the $E_{2.88}$ models had statistically significant predictive power for E . Yet, the

predictive power of the models decreased in early and late season for apple trees and the late season for pear trees. In apple trees, canopy conductance during the early and late season was significantly higher than mid-season which was also reflected in a greater decoupling of canopy conductance from atmospheric evaporative demand and lower predictive power of the $E_{2.88}$ models. Conversely, in pear trees late season canopy conductance was significantly lower from early and mid-season because of a tighter coupling of biophysical constraints on canopy conductance. Overall, this study found that a consideration of phenology, particularly for deciduous species, is required to improve the accuracy of $E_{2.88}$ models.

The decoupling factor (Ω) was significantly greater in early and late season, compared with mid-season, for the apple trees indicating that canopy conductance was less constrained by atmospheric evaporative demand. In contrast, pear canopy conductance and Ω were significantly

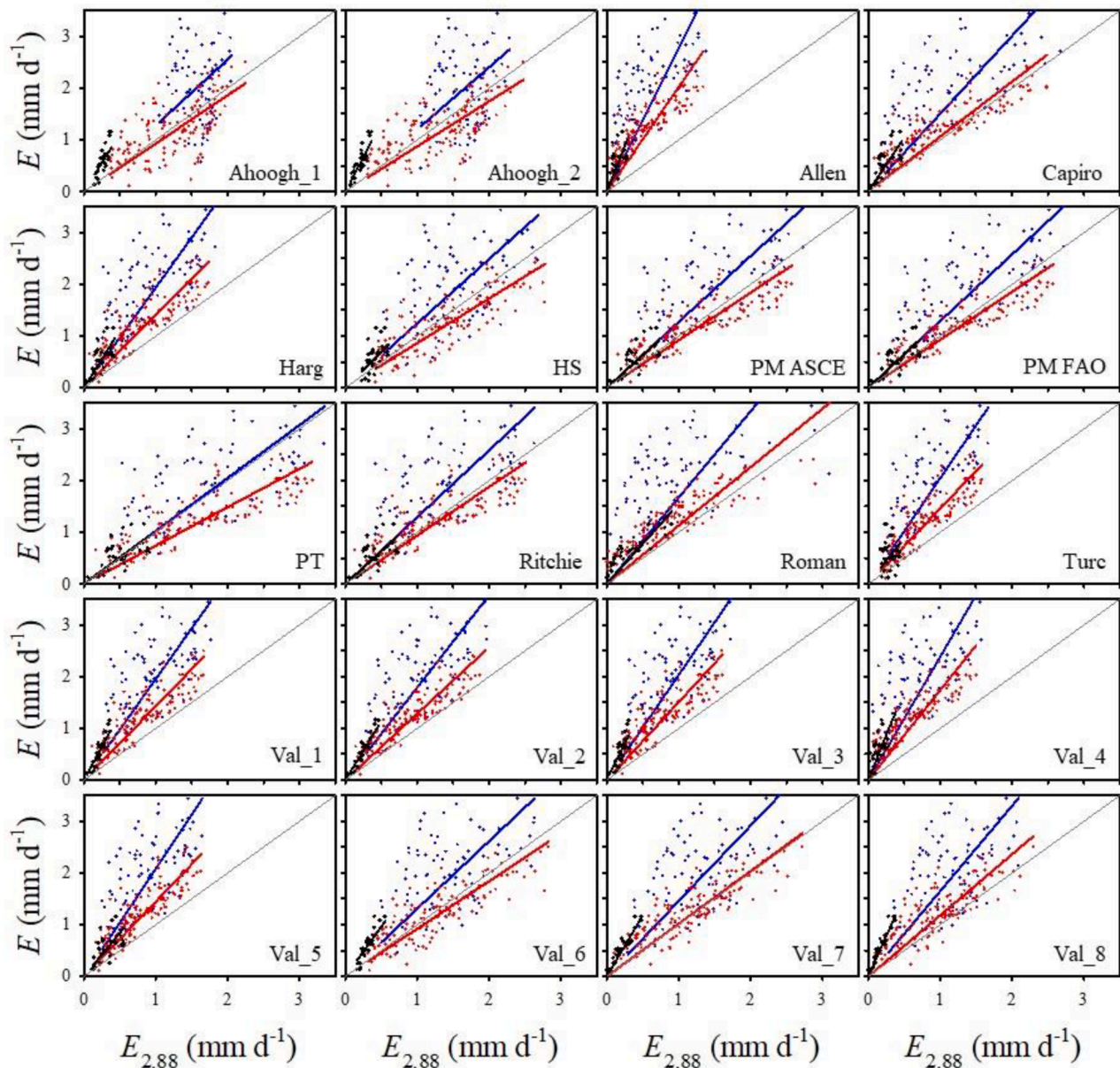


Fig. 4. The influence of phenology on the predictive power of 20 $E_{2.88}$ models for apple tree E during the early (blue), mid- (red) and late (black) season. The $E_{2.88}$ model abbreviation is labelled on each graph and defined in Table 2. The grey line is the 1:1 line of unit. Statistical results are presented in Tables 3 and 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

lower in the late season indicating that canopy conductance was increasingly coupled with atmospheric humidity and temperature. The Ω for apple ranged from 0.65 in the early/late season to 0.54 for the mid-season. For pear, Ω ranged from 0.54 to 0.48. Previous studies observed $\Omega < 0.2$ in apple (Cohen and Naor, 2002), 0.12 to 0.30 in grapevine (Lu et al., 2003), and 0.09, 0.11 and 0.22 for coffee, citrus and sugarcane, respectively (Marin and Angelocci, 2016). These authors concluded that canopy conductance was strongly coupled with atmospheric evaporative demand because of the low value of Ω . Greater values of Ω , canopy conductance and sap flow have been observed in juvenile versus mature foliage (Phillips et al., 2010). Similarly, stomatal conductance in younger leaves was higher than mature leaves in potted sapling apple trees (Thorpe et al., 1980). In contrast, Nabi et al. (2000) found greater rates of stomatal conductance in mature leaves versus juvenile leaves along single branches in apple trees. The observed differences between apple and pear in this study may have been caused by difference in xylem vessel diameter, leading to changes in root and stem hydraulic

conductance, that has previously been observed for these species (Atkinson et al., 2003; Jupa et al., 2021). For pear trees, the late season decrease may have been caused by a decrease in stem and leaf hydraulic conductance often observed in senescing leaves (Brodribb and Holbrook, 2003). A greater decoupling of canopy conductance with atmospheric evaporative demand may have been caused by low stomatal control of transpiration or xylem vessel formation. Although it was beyond the scope of the current research, future studies can test these possible causes by measuring stomatal pore aperture, stomatal conductance, xylem vessel diameter, and/or xylem hydraulic conductivity.

A second major outcome of this study was comparing sap flow, or transpiration, against a variety of ET_o models. Many studies compared the performance of a variety of ET_o models against the PM FAO model (e.g. Djaman et al., 2015, 2019; Ndulue and Ranjan, 2021; Song et al., 2019; Júnior et al., 2020) or independent instrumentation such as lysimeters (Liu et al., 2017), eddy covariance (Babatunde et al., 2017; Ha

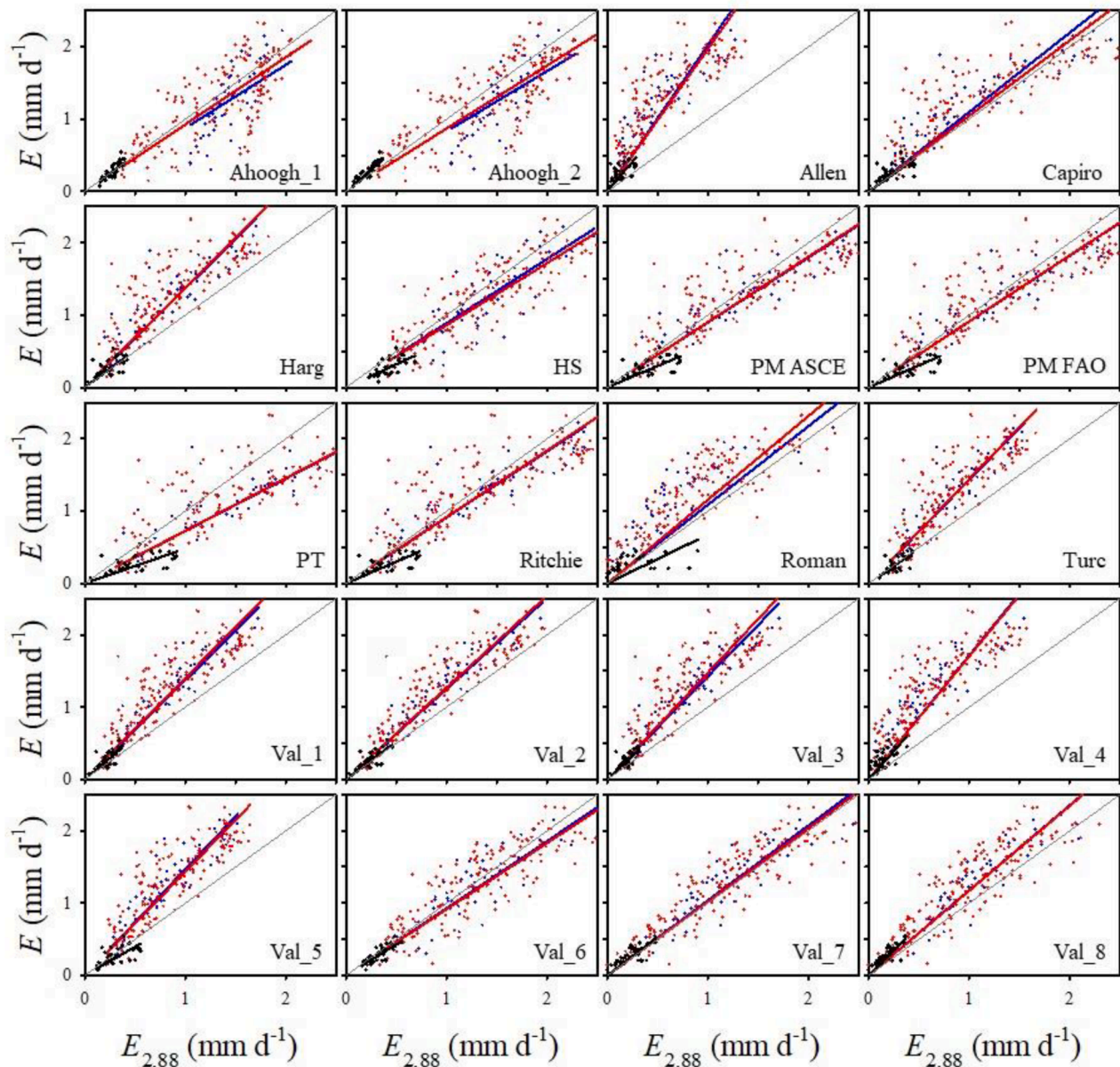


Fig. 5. The influence of phenology on the predictive power of 20 $E_{2.88}$ models for pear tree E during the early (blue), mid- (red) and late (black) season. The 20 $E_{2.88}$ model abbreviations are labelled on each graph and correlate with those defined in Table 2. The grey line is the 1:1 line of unit. Statistical results are presented in Tables 3 and 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

et al., 2015), or scintillometry (Minacapilli et al., 2015). In contrast, sap flow is mainly compared against PM FAO and rarely against other ET_o models (e.g., Ayyoub et al., 2017; Cammalleri et al., 2013; Caspari et al., 1993; Forster and Englefield, 2021). It was previously demonstrated that E was highly correlated with the $E_{2.88}$ model derived from PM FAO (Forster, 2021) and this result was further supported by observations in apple and pear. This study advanced this finding by demonstrating that E is additionally highly correlated with several other ET_o models when those models are also normalized by the hypothetical reference grass leaf area index of 2.88.

The PM FAO and PM ASCE models had strong predictive power with errors of less than 7% when $E_{2.88}$ was compared with E during the mid-season. The strong performance of the $E_{2.88}$ model derived from PM FAO and PM ASCE was supported by previous studies (Ayyoub et al., 2017; Fernández et al., 2008; Forster, 2021; Pereira et al., 2006, 2009). However, the Val7 model (Valiantzas, 2018) had the strongest predictive power with errors of 2% for apple and pear during the mid-season.

The Val7 model is a simplification of the PM FAO that does not require solar radiation or wind speed and may prove advantageous in future research comparing sap flow against ET_o across a broader range of species. Other models examined in this study had mixed performance and, generally, the weakest models featured fewer parameters and meteorological inputs than PM FAO 56, PM ASCE or Val7 models, an outcome supported by previous comparisons of ET_o model performance (Djaman et al., 2017). Additionally, RH was a feature of the models with more predictive power yet was not included in the weakest performing models (e.g., Allen, Harg, Turc, Val4). It is well known that RH (and D) are strongly correlated with Q and E across a wide range of species and vegetation types (e.g., David et al., 2004; Forster, 2012; Forster and Englefield, 2021; Granier et al., 1996; Oren et al., 1999; Paudel et al., 2015). Therefore, future research considering simplified ET_o models or instrumentation should include RH for more accurate correlation between Q and ET_o .

The measurement of Q and A_L may also provide an additional source

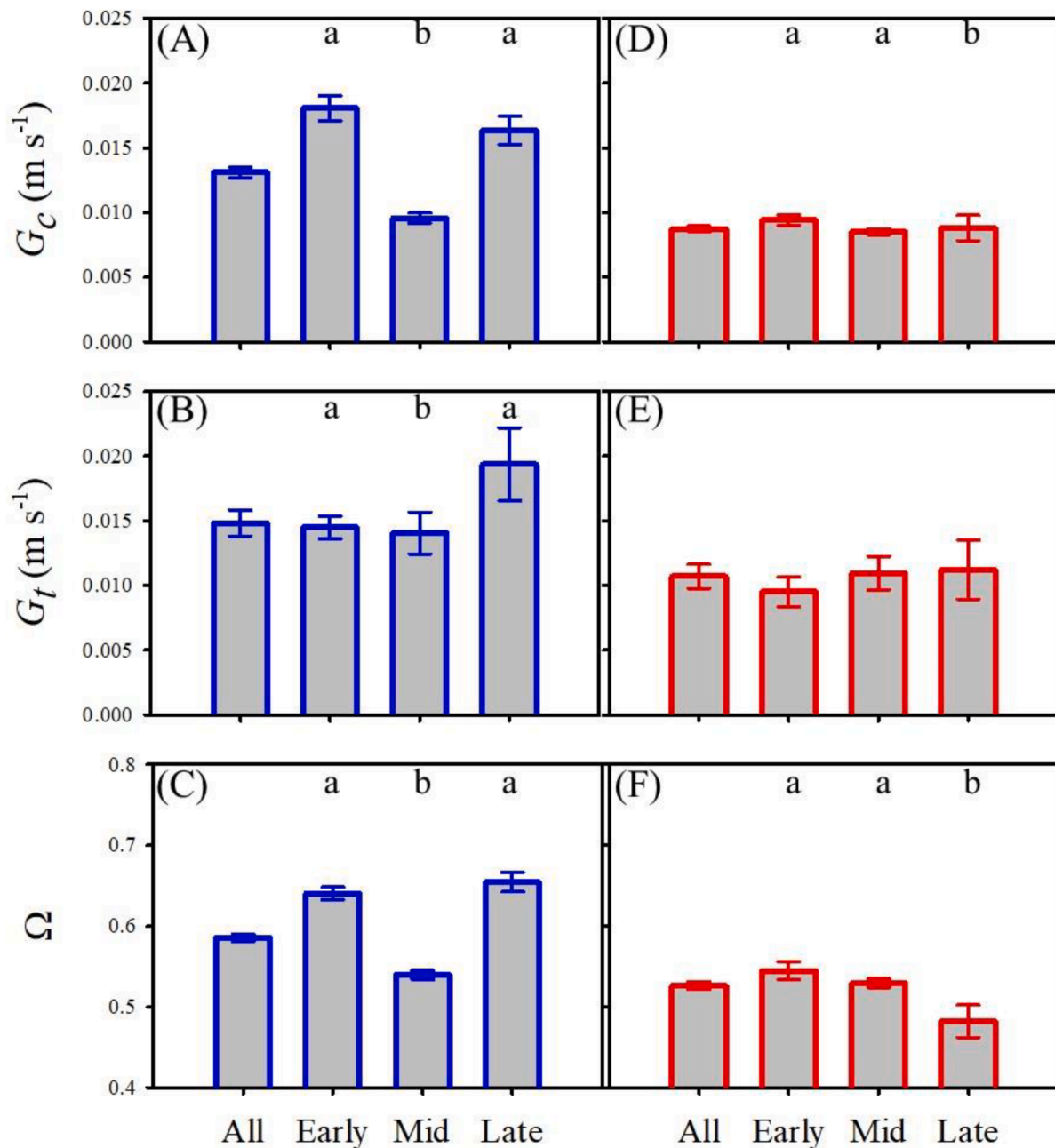


Fig. 6. The mean $\pm$ standard error of canopy conductance (G_c), conductance of the ‘average’ stomata (G_t) and the decoupling factor (Ω) for apple (blue, A, B, C) and pear (red, D, E, F) for all, early, mid- and late season. The Tukey HSD post hoc results of the one-way ANOVA between early, mid- and late season are indicated by lowercase letters above the bars. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

of uncertainty in the calculation of E . For example, a review found that Q had a bias towards the underestimation of E by 34.7 % (Forster, 2017). The bias towards underestimation is caused by various independent and correlated parameters including wound effects, probe misalignment, sapwood moisture content, and the measurement range limitation of sap flow methods based on principles of thermal conduction and convection (Caspari et al., 1993; Čermák et al., 2004; Forster, 2017). These sources of error were carefully corrected in this study via accurate measurements of wounding, probe misalignment and sapwood water content. Furthermore, the dual method approach (DMA) resolves the measurement range limitation of thermometric sap flow sensors (Forster, 2019, 2020). The DMA has also accurately estimated Q in an urban weed species (Forster, 2019), 15 woody species (Forster, 2020), pear trees (Forster, 2021), and grapevines (Forster and Englefield, 2021). In this study, maximum E for apple was approximately 3.6 mm d^{-1} compared

with 2.1 mm d^{-1} observed in dwarf apple trees in New York, USA (Dragoni et al., 2005) and 1.5 mm d^{-1} for a controlled irrigated ‘Braeburn’ apple in Palmerston North, New Zealand (Mpelasoka et al., 2001). The accurate measurement of A_L depends on the accurate measurement of leaf area index (LAI) and canopy projection area. In this study, two independent measurements of LAI were employed that showed the same results. The accuracy of the light interception method of estimating LAI has previously been demonstrated in several studies (Fang et al., 2019; Garrigues et al., 2008). The accuracy of the digital photograph algorithm also used in this study was previously demonstrated by Fuentes et al. (2008). In this study, observed maximum apple LAI was 2.9 compared with a LAI of 1.7 to 3.9 for various apple varieties grown around the world (Cohen and Naor, 2002; Liu et al., 2012, 2015; Mobe et al., 2020; Tustin et al., 2001). Canopy projection area was carefully measured in this study. But this parameter is known to be difficult to

resolve precisely due to the canopy coarseness, canopy shape, scale of measurement, and other factors (Seidl et al., 2011; Zhu et al., 2020). Other biometric parameters may be considered as easier to measure and more accurate than LAI and canopy projection area (Čermák, 1989) and their suitability for the $E_{2.88}$ or similar can be explored.

5. Conclusion

In this study, a comparison of 20 $E_{2.88}$ models against apple and pear transpiration revealed novel findings including: i) that $E_{2.88}$ models generally have good predictive power for apple and pear transpiration during the mid-season; ii) yet, the predictive power of all models decreased during the early and late season due to a decoupling of canopy conductance with atmospheric evaporative demand; and iii) $E_{2.88}$ models that incorporated a humidity parameter had greater predictive power than models that did not. For apple, the Penman-Monteith ASCE-EWRI model had the highest predictive power, whereas for pear the Valiantzas (2018, Eq. (7)) was the most accurate. Future studies should investigate the physiological and morphological factors that contribute to the decoupling of canopy conductance with atmospheric evaporative demand and how $E_{2.88}$ models can be improved by the incorporation of dynamic vegetation parameters.

Declaration of Competing Interest

MAF is an owner of a commercial company that manufactures sap flow sensors that were used as part of this study. The following authors have no conflict of interest to declare: TDHK, SK, MA, VRC, JS, RNM.

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